SERVER:
OPEN CHIPLET
ECONOMY

Agile Silicon. Will Open Chiplet Architectures Keep Pace with Fast-Evolving AI Dataflows

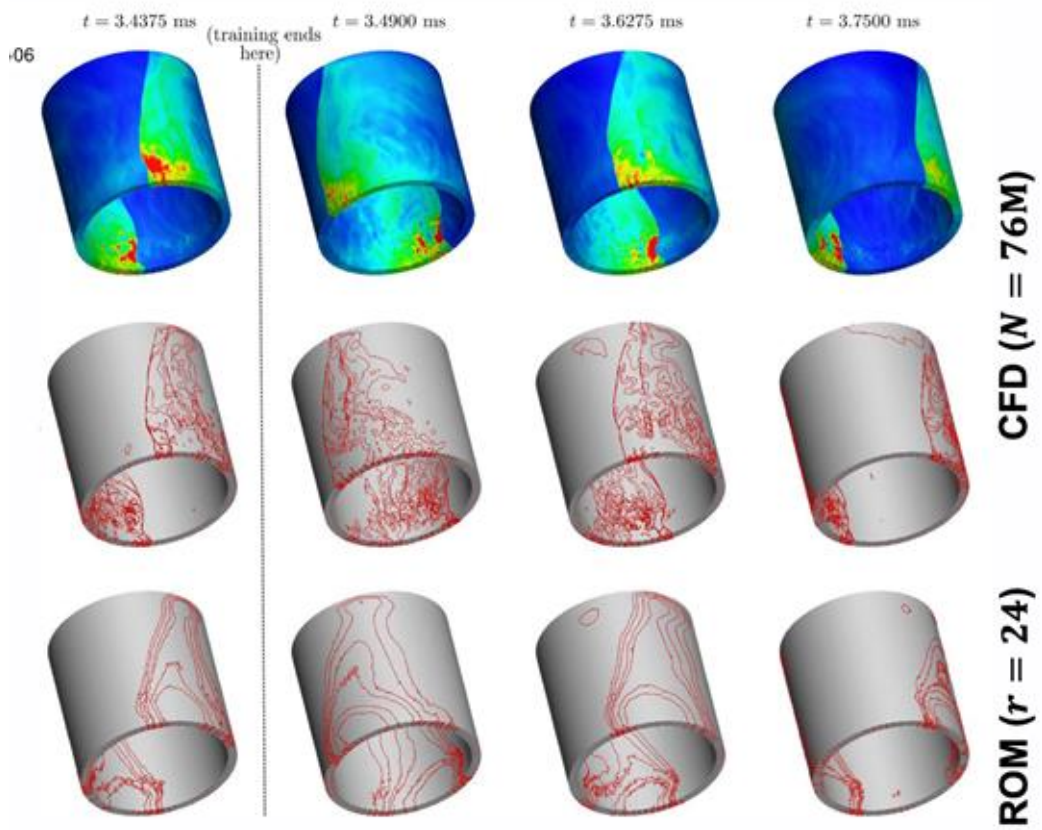
Patricia Gonzalez-Guerrero

Lawrence Berkeley National Laboratory

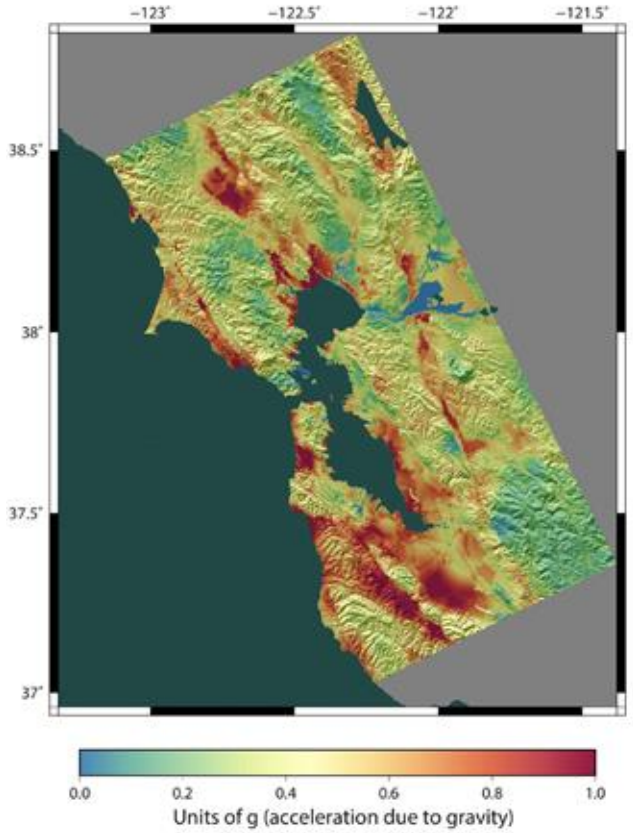
On behalf of the chiplets modularity for AI/HPC workstream

OCP Alliance Partner

In our minds HPC is Equivalent to Scientific Computing



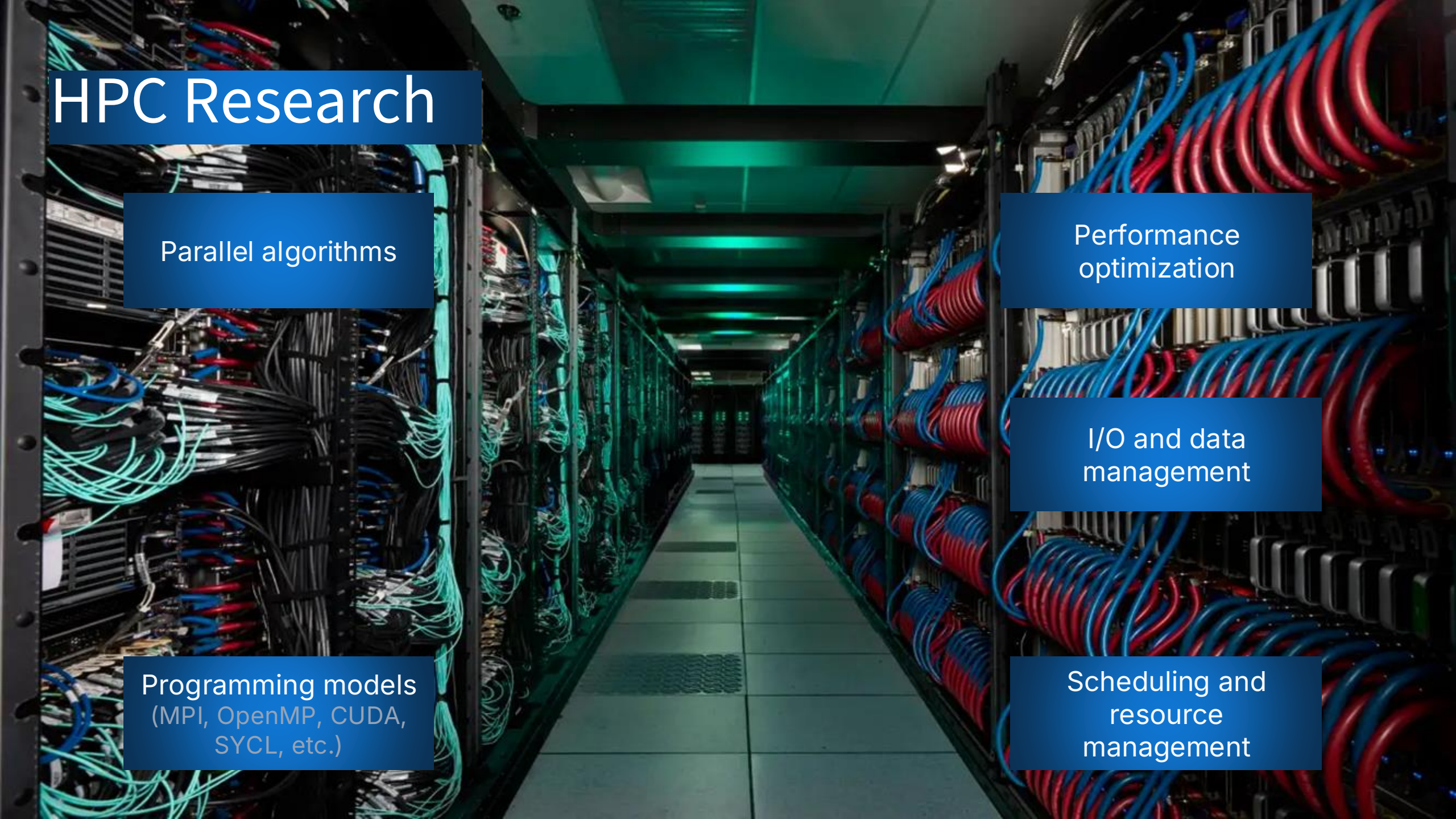
Karen Willcox

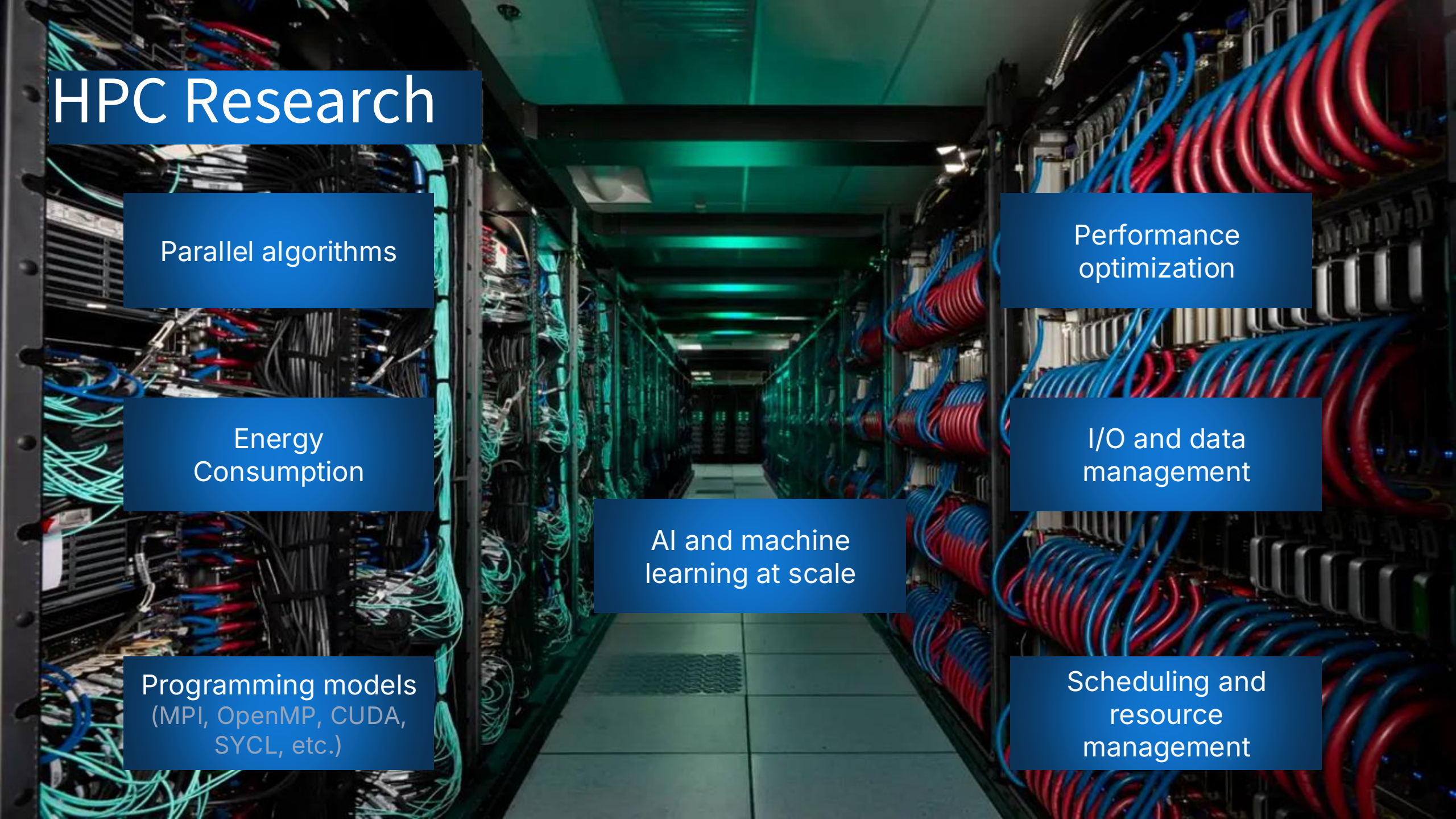


Ewa Deelman

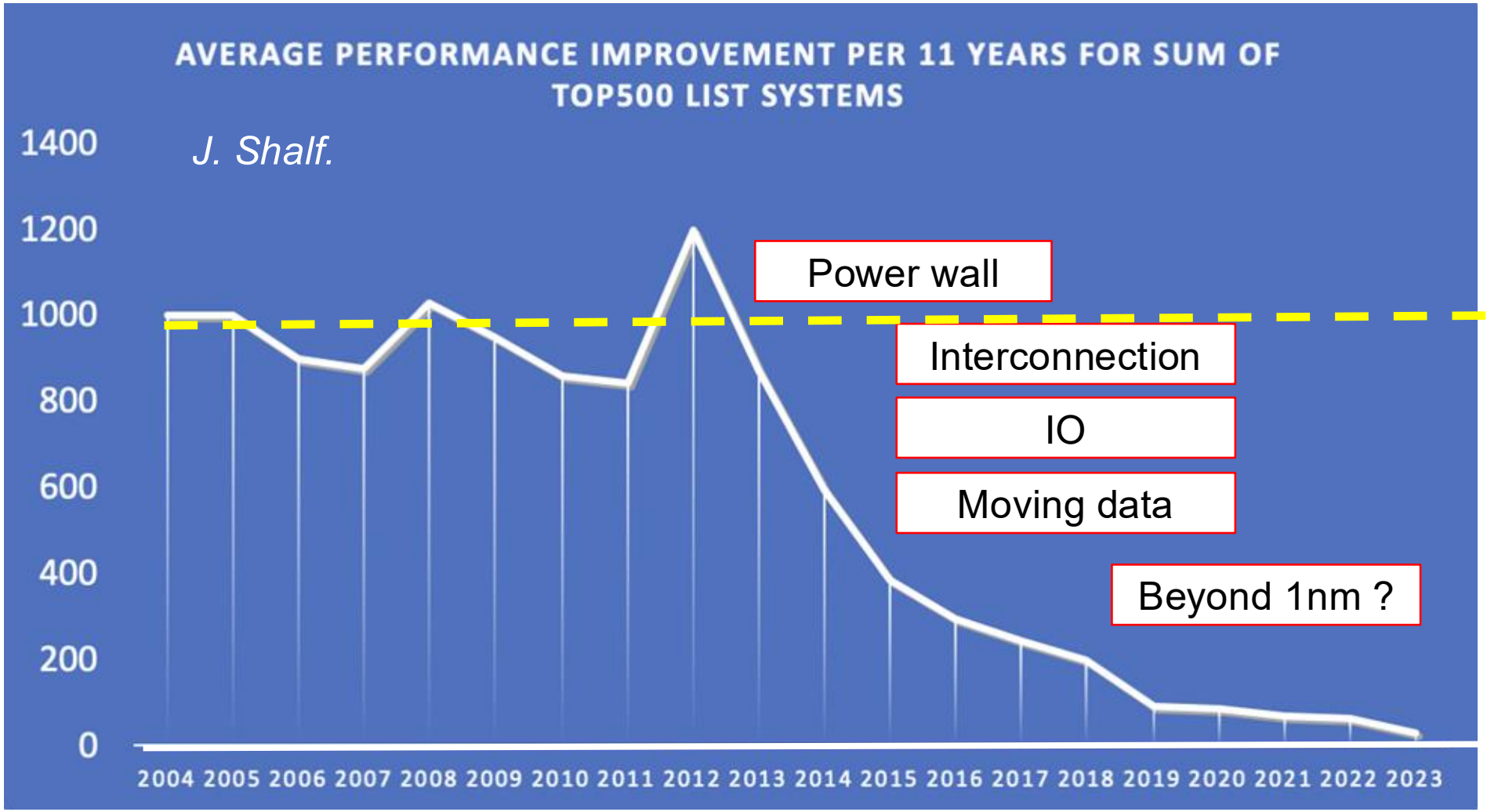
4130 nodes, running for **45 days**, producing/ingesting **1PB data**. **55K node-hours**

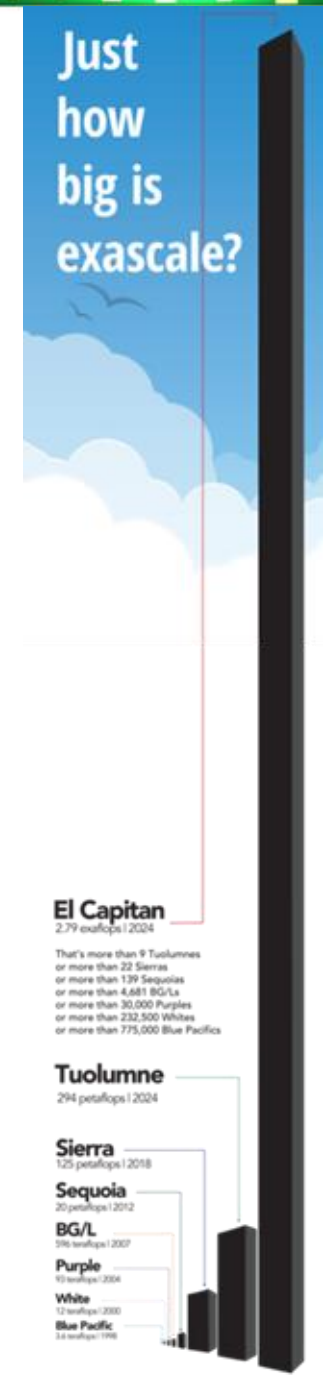
Ilkay Altintas


HPC Research
Parallel algorithms
Performance optimization
I/O and data management
Programming models (MPI, OpenMP, CUDA, SYCL, etc.)
Scheduling and resource management

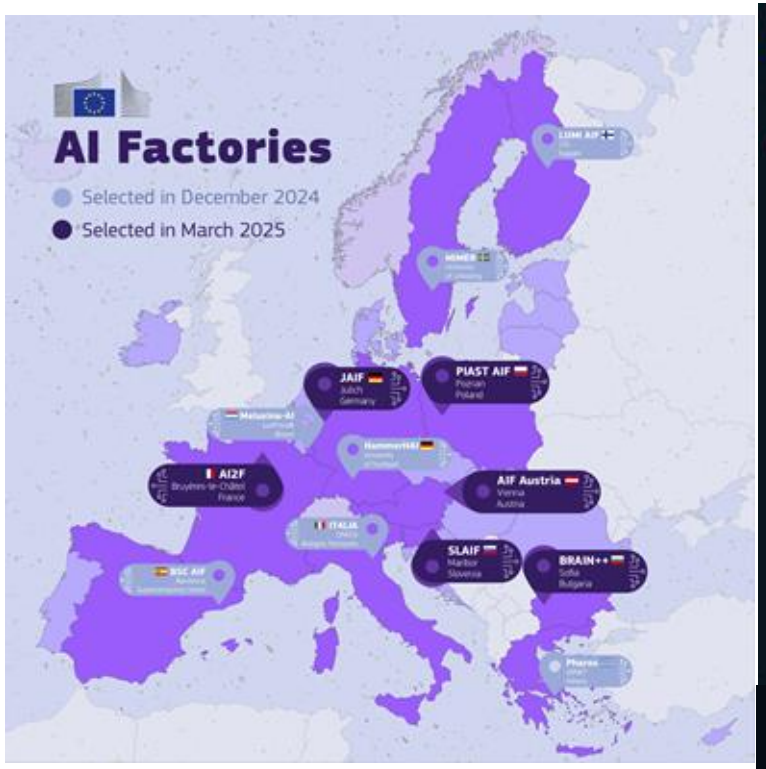

HPC Research
Parallel algorithms
Performance optimization
Energy Consumption
I/O and data management
AI and machine learning at scale
Programming models (MPI, OpenMP, CUDA, SYCL, etc.)
Scheduling and resource management

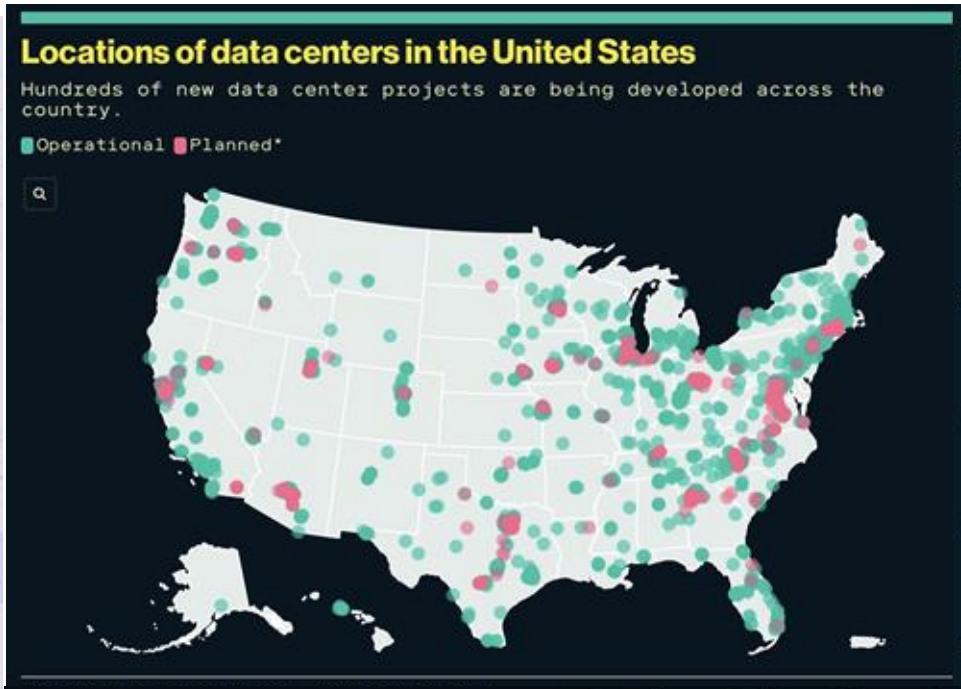
Performance is not increasing any more

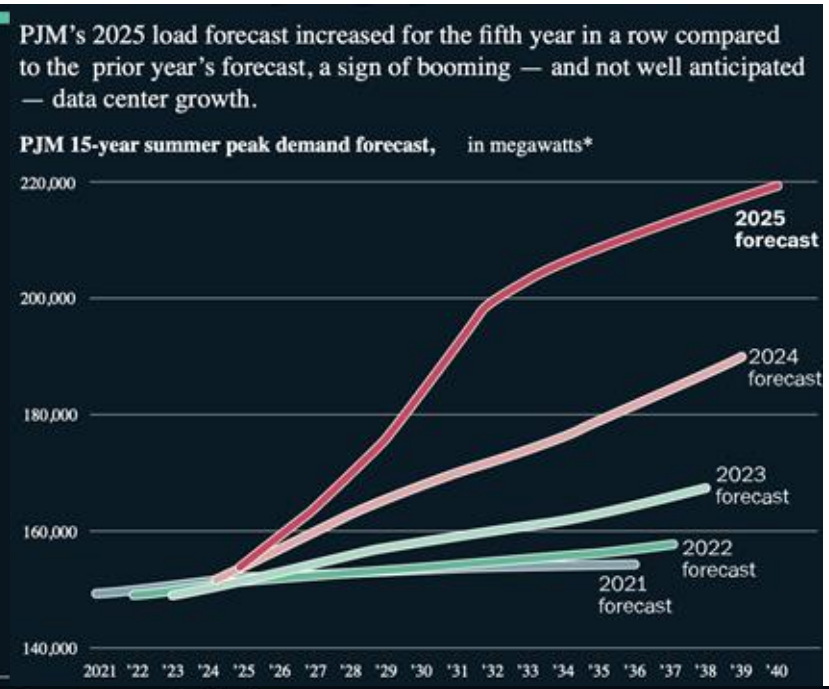




On Top of Performance, a Looming Energy Crisis







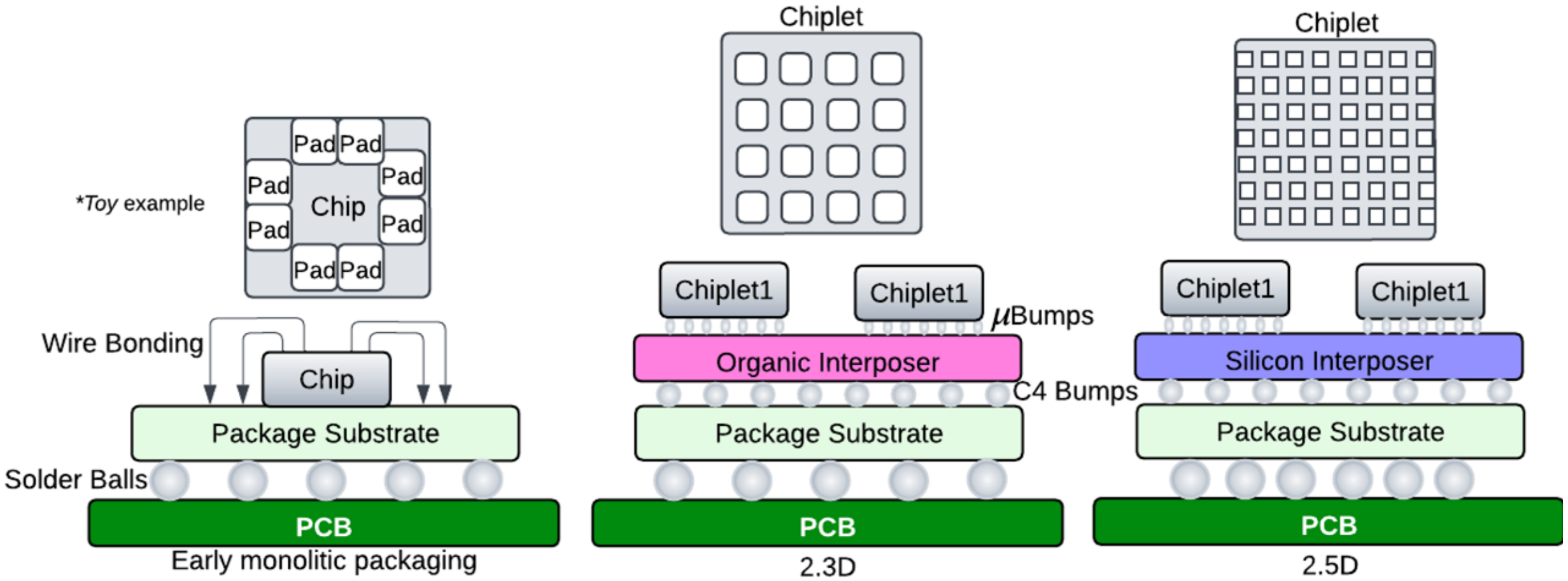
Giga Watt AI factories!

Series: The New Energy Crisis

https://thenewenergycrisis.com/

An investigative series uncovering how Virginia's data center boom—fueled by AI—threatens to overwhelm the power grid, drive up household energy costs, and put entire communities at risk.

Advanced Packaging & Chiplets



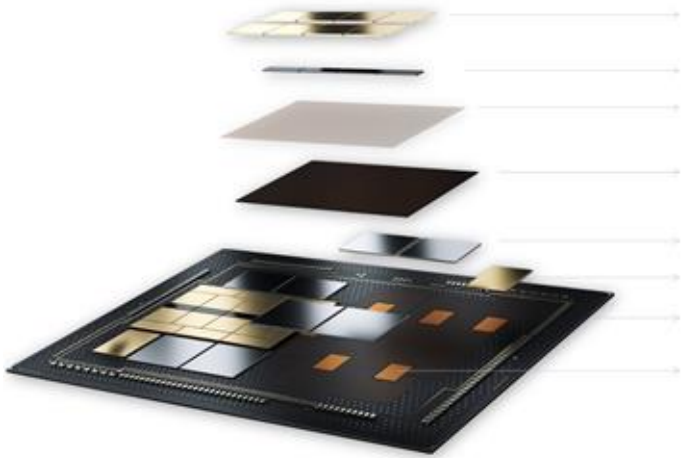
Chiplets made Exascale possible*

**And other key technologies: Novel interconnects, advanced cooling, power delivery… :-)*

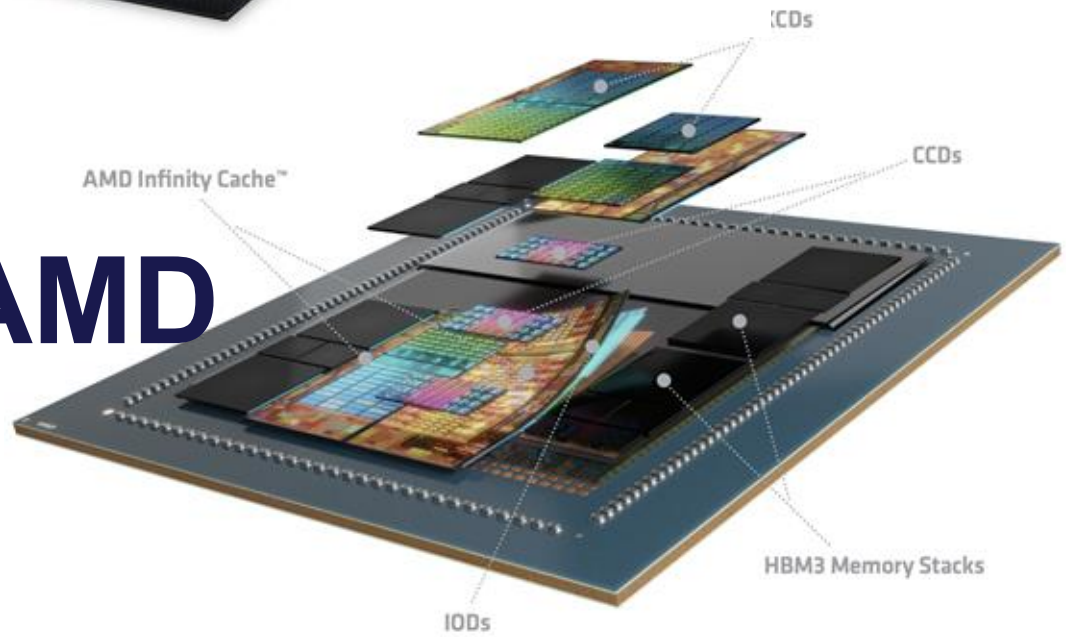








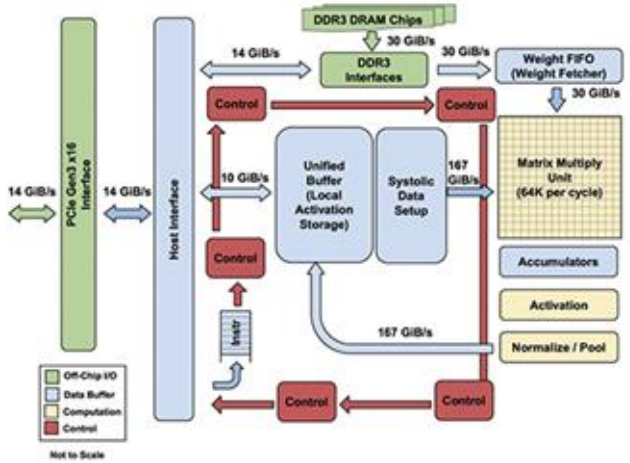
Intel



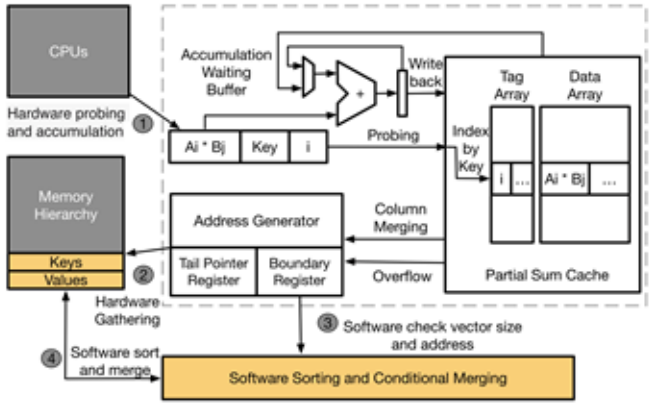
AMD



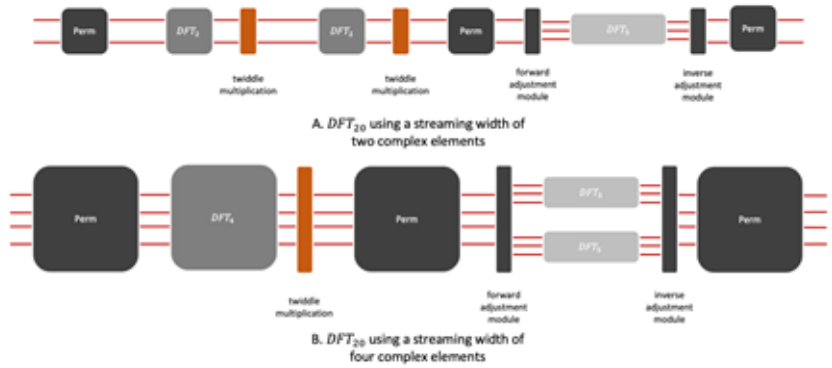
What is next: Specialization!



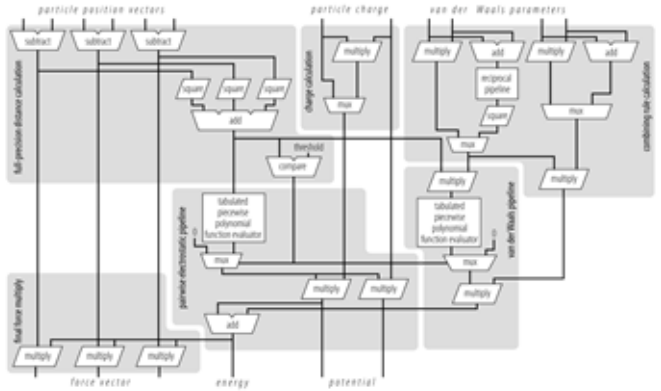
Tensor Processing Unit for Matrix-Matrix Multiplication. Google



Sparse Accumulation for SpGEMM. Lehigh University and LBNL



Hardware Implementation for Mixed-Radix Fourier Transform. LBNL



Pairwise Point Interaction Pipeline for Molecular Dynamics. Anton.

Potential

- 100x - 1000x more performance
- 100x - 1000x less energy consumption

Challenges

- Overspecialization
- Algorithms evolve faster than hardware
  - RNN

How do we leverage chiplets modularity so hardware specialization makes it into HPC clusters?



AMD. Two different products.
Same chiplets => modularity!!

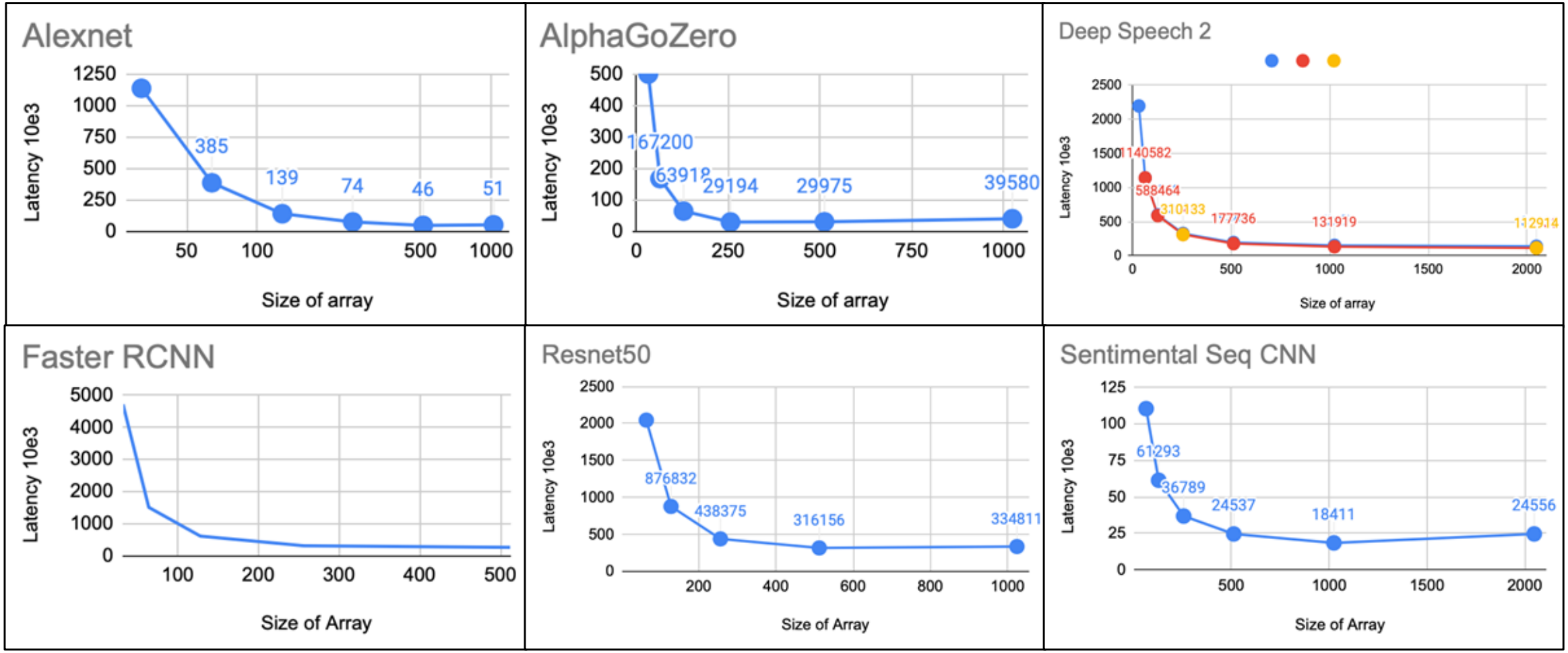
Four Steps to Specialization

1. *Profiling algorithms: Find bottlenecks*
2. *Macroarchitecture*
   a. *Chiplets*
   b. *Interposer*
3. *Chiplet parameters*
   a. *Form factors, interfaces,*
4. *Micro/Macro architecture*

Chiplets Marketplace ;)

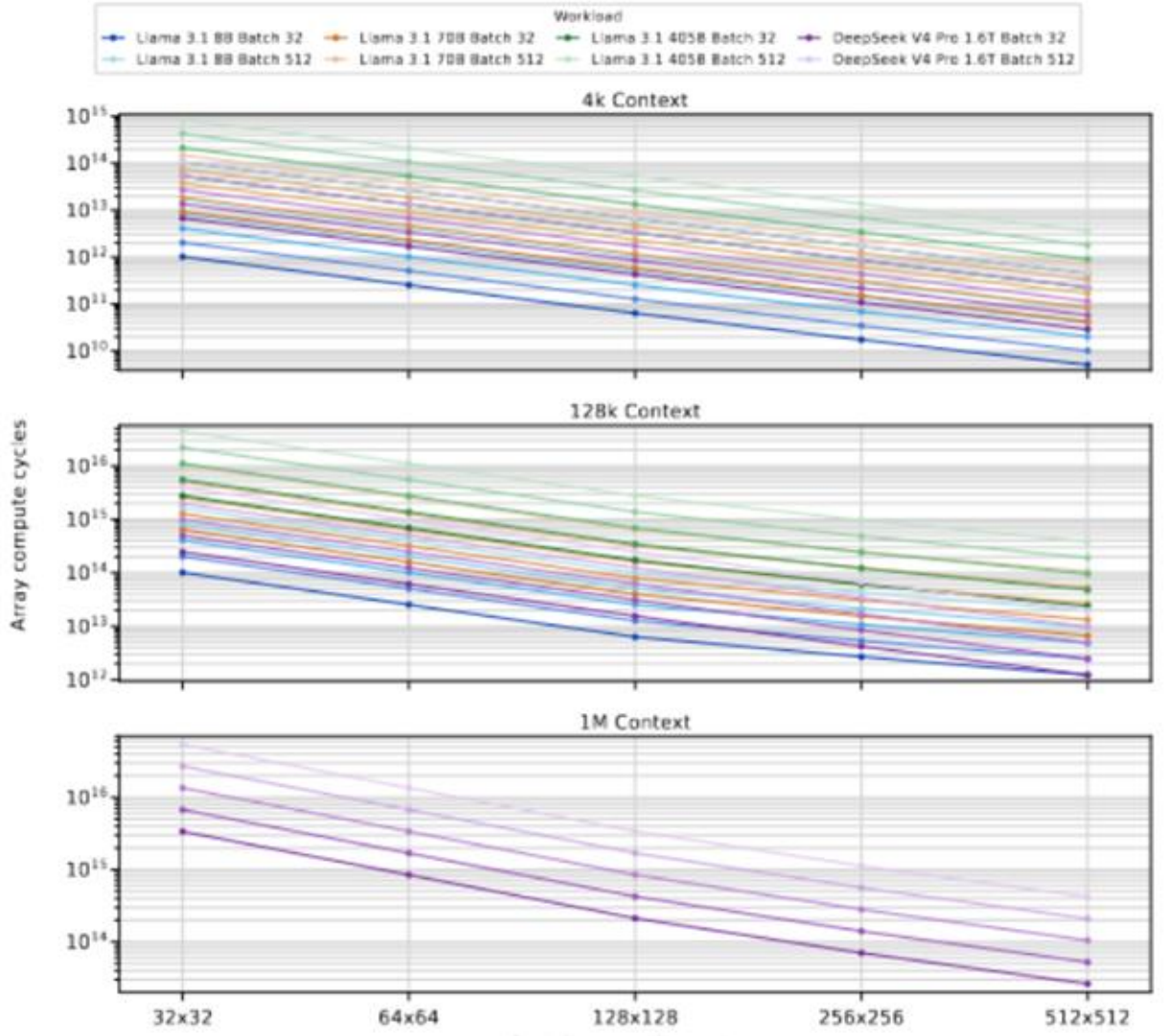


S1: Profiling algorithms - Machine learning



Hypothetical systolic array

S1: Profiling algorithms - Large Language Models

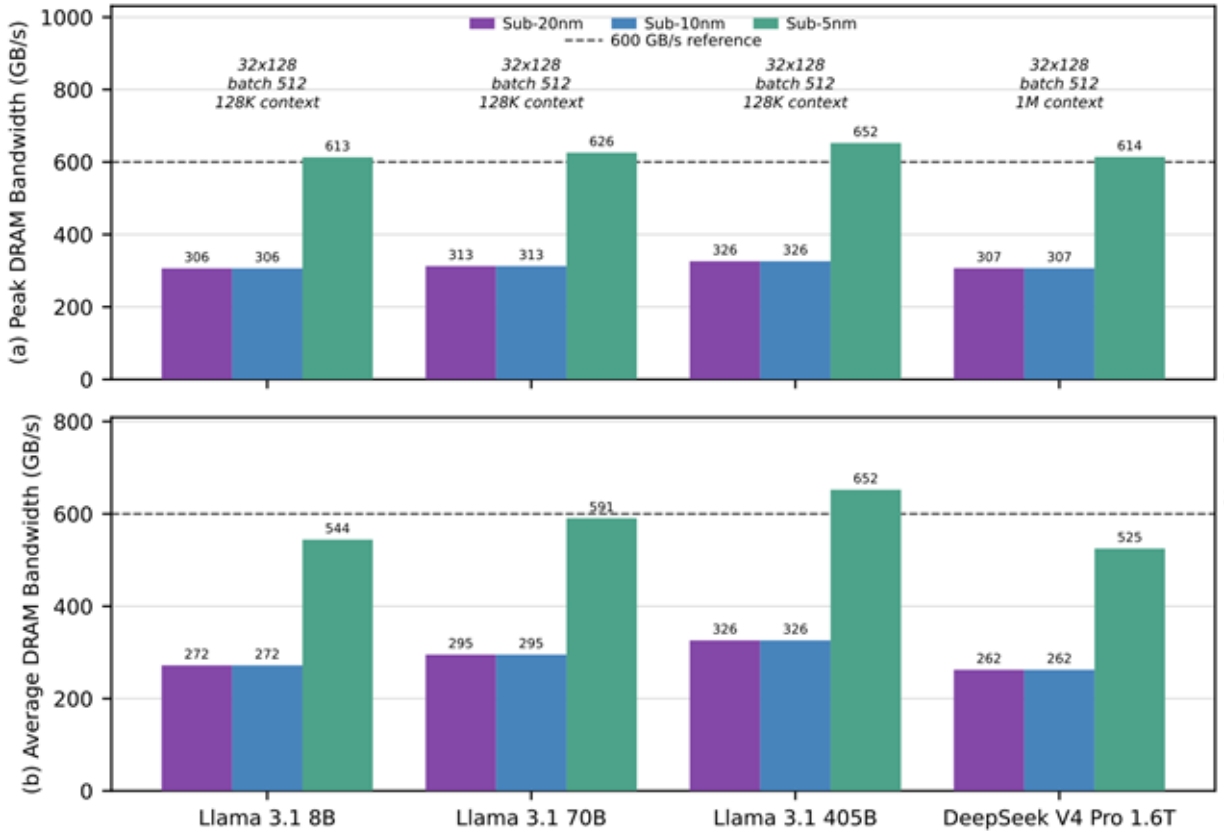


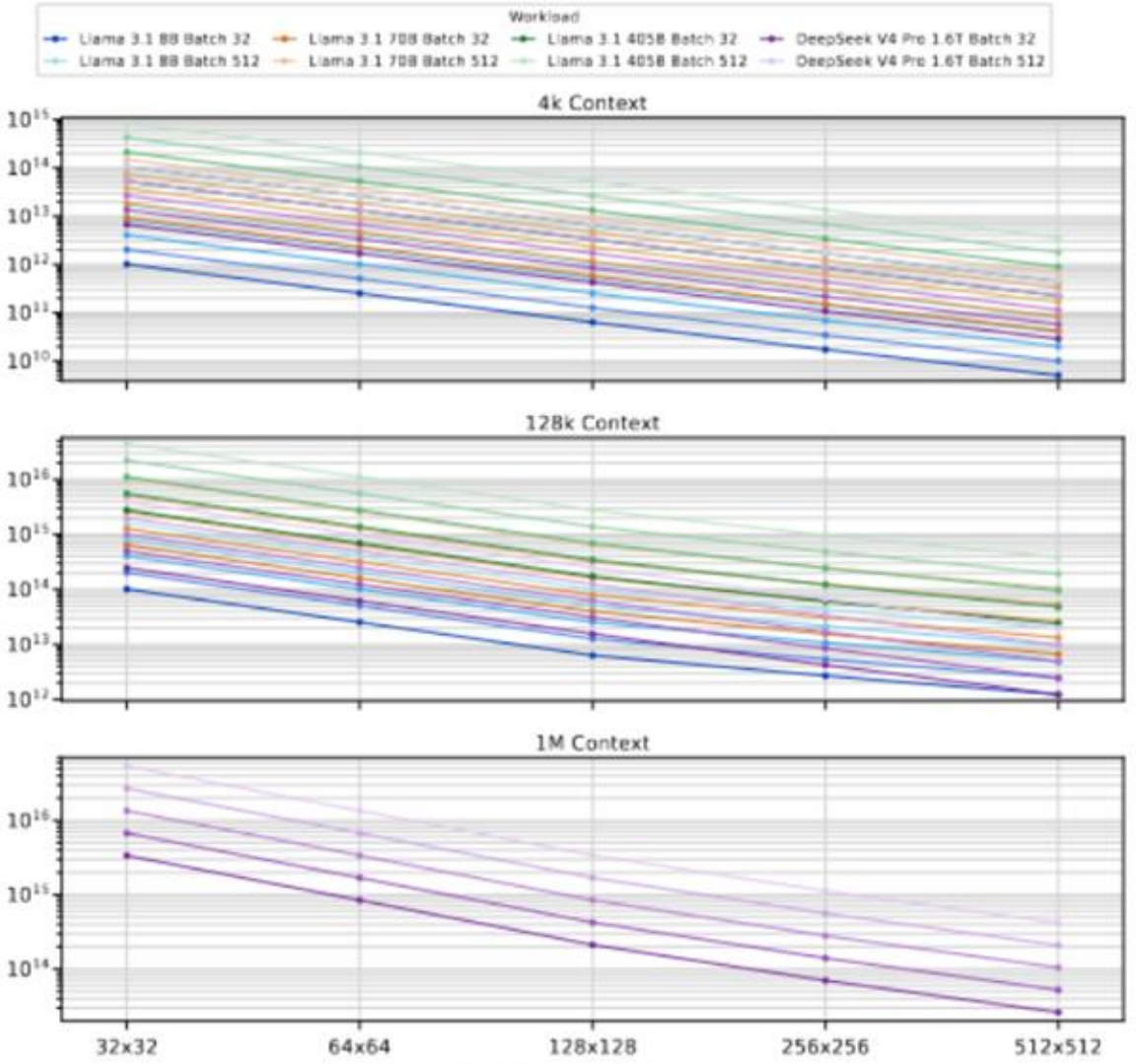


Hypothetical systolic array

University of Central Florida

S1: Profiling algorithms - Large Language Models





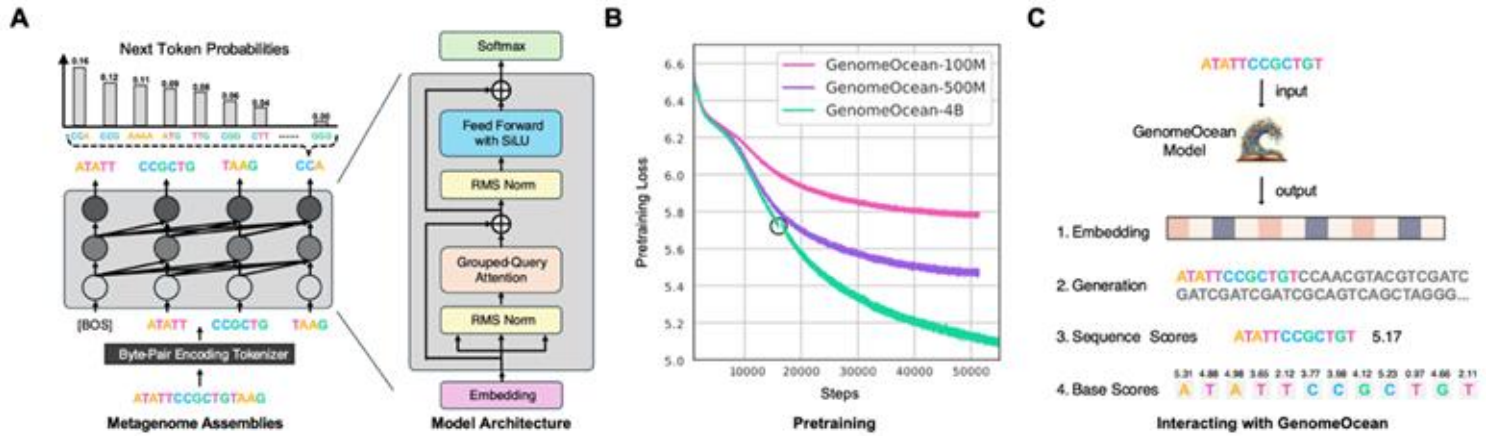


Hypothetical systolic array

University of Central Florida

S1: Profiling algorithms - Specialized LLMs

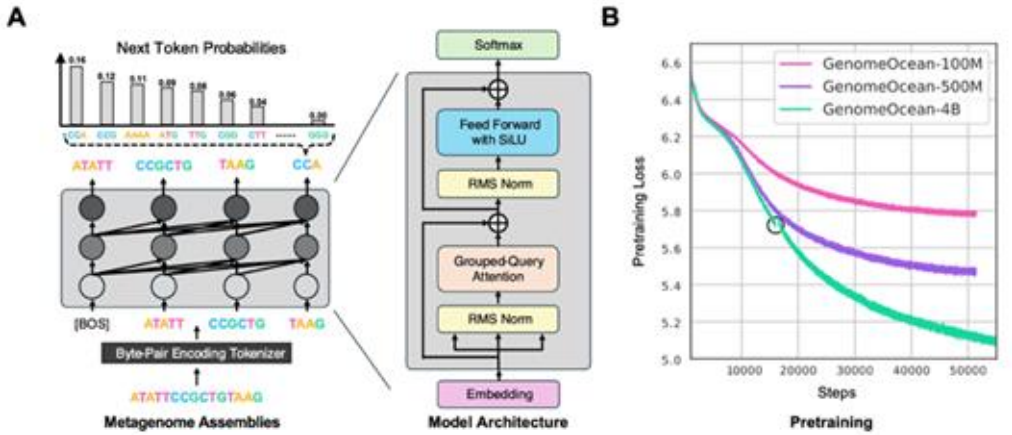
GenomeOcean: An Efficient Genome Foundation Model Trained on Large-Scale Metagenomic Assemblies





S1: Profiling algorithms - Specialized LLMs

GenomeOcean: An Efficient Genome Foundation Model Trained on Large-Scale Metagenomic Assemblies



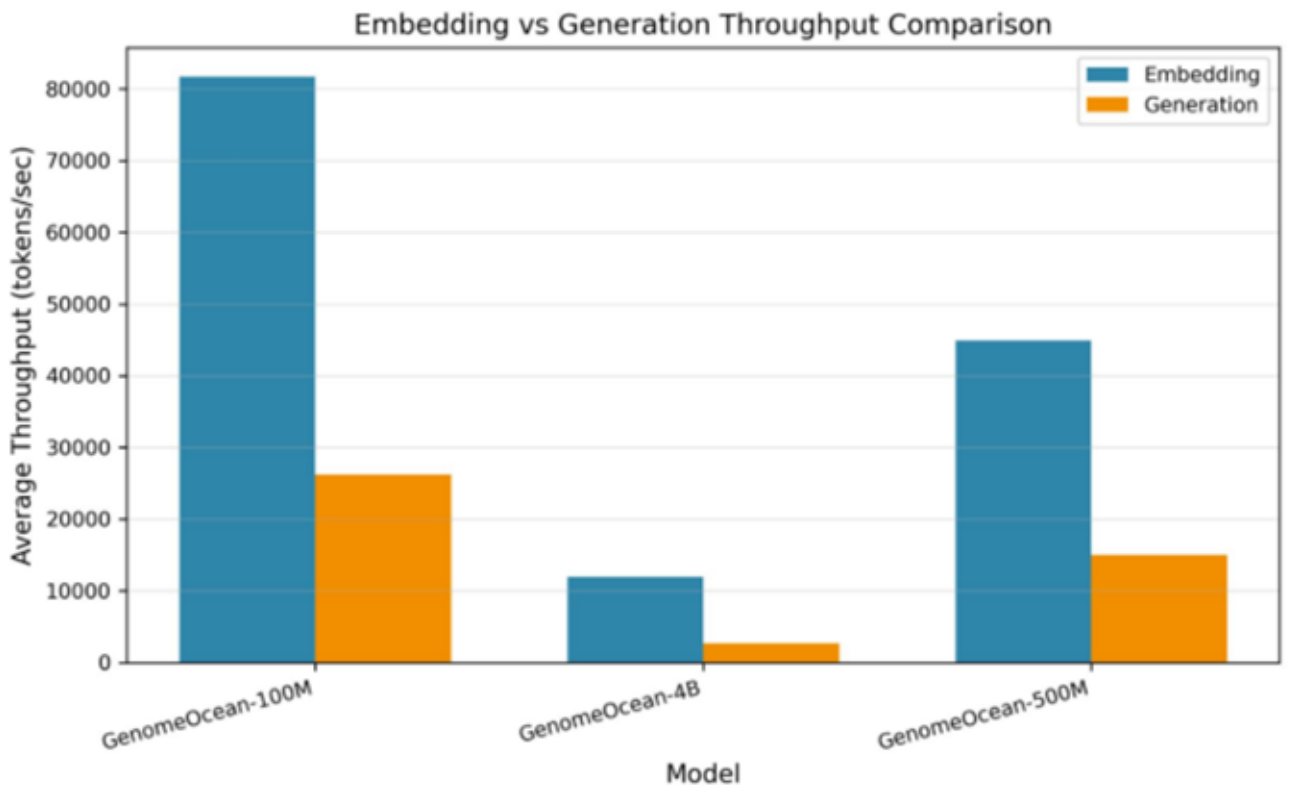
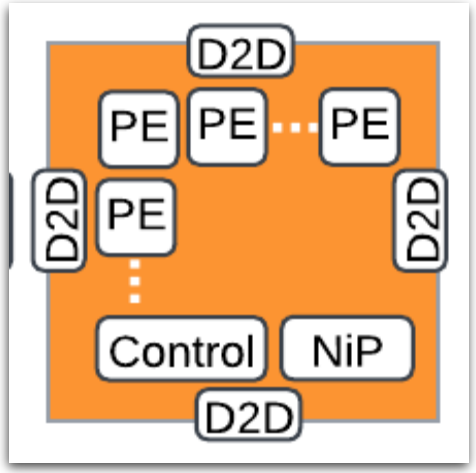


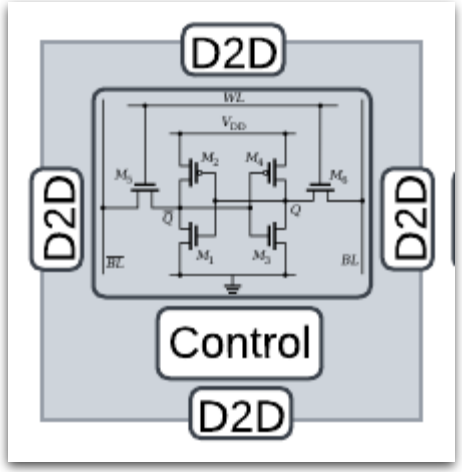
Figure 1: Embedding vs Generation average throughput comparison across all three models.



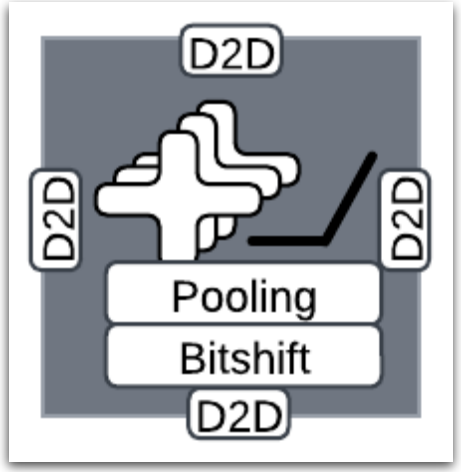
S2: Chipletization of a specialized hardware accelerator



Systolic Array Chiplet
Grid of PxP PE



Scratchpad Chiplet
SRAM based



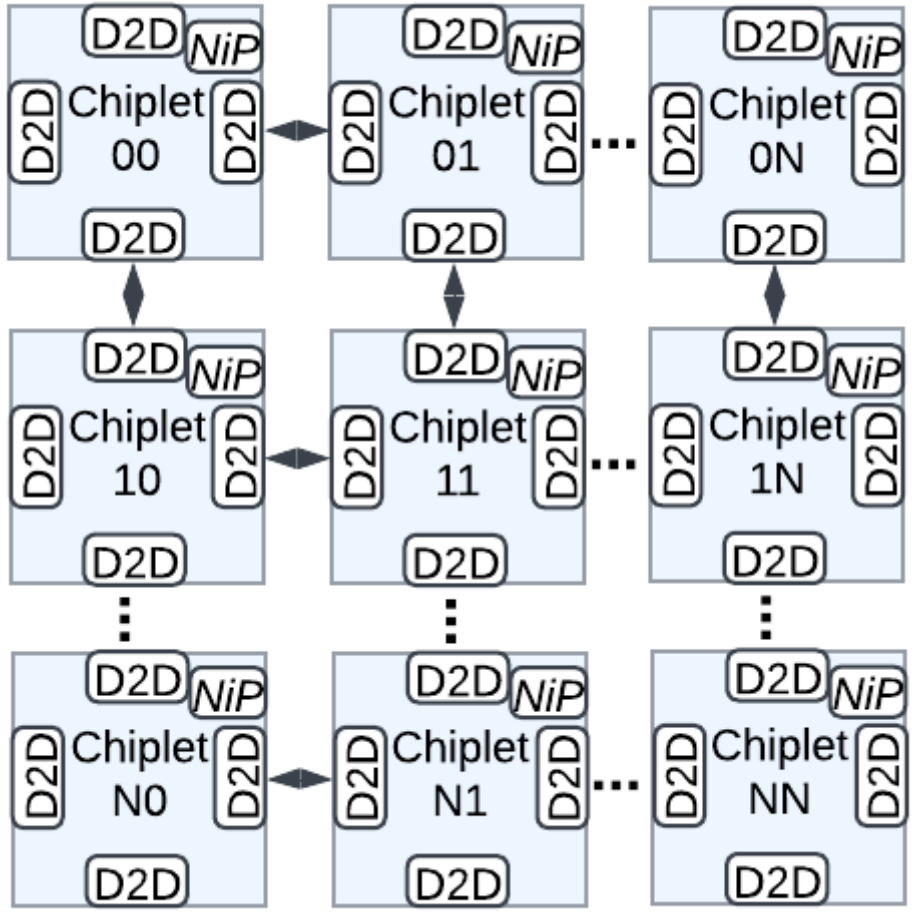
Accumulator Chiplet

Chiplets Marketplace ;)



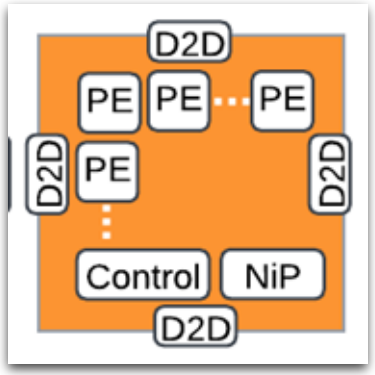
S2: Scalable Interposer 2D grid with NxN Chiplets

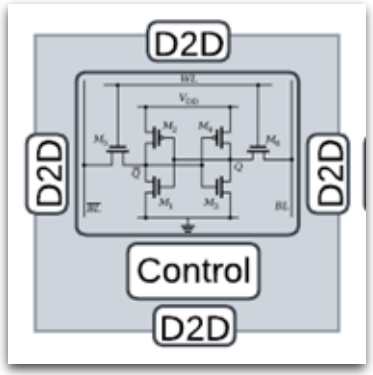
- **Network-In-Package (NiP)**
  - Packet switched
  - Light weight
  - Minimum transmission latency (1CC)
- **No Cache Coherency**
  - Message passing for parallel computing
    - Hardware Message Queues

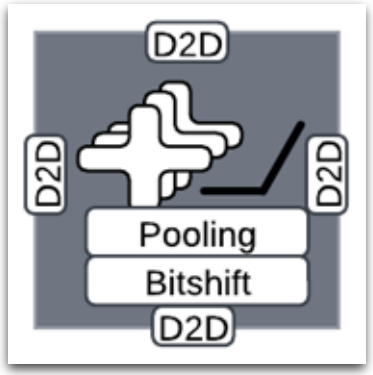


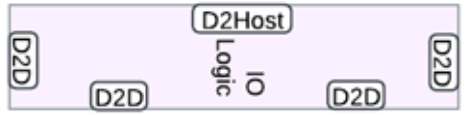
Dataflow is regular and predictable

S3: Some math to define our chiplets parameters









	Chiplet 4 mm x 4mm	Sub 20nm @ 1GHz	Sub 10nm @ 1GHz Sub 5nm @ 2GHz
1	**Systolic Array**	64 x 64 PEs	128 x 128 PEs
2	**Scratchpad**	1MB	9MB
3	**Accumulator**	64 acc	128 acc

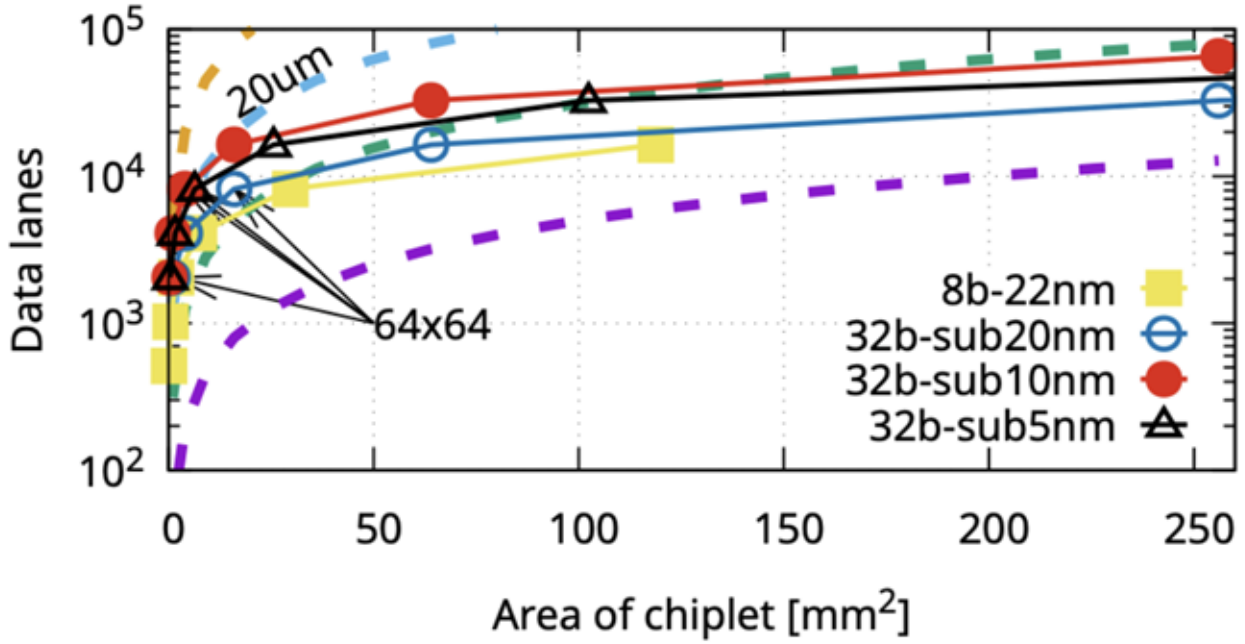
S3: Some math to define our chiplets parameters

Bump Pitch = 20um

- Chiplet area 4mm x 4mm
- Oversized Buffers

Bump Pitch > 40um

- Standard Interfaces: UCIe, BoW
- High frequency + serialization = more Power and latency



Area of chiplet: Logic (Systolic array) + NiP + D2D Interface

S4: Three different SiPs for three dataflows

	Sub 20nm SiP		
	CNN	Deep Speech	Trans.
Syst. Array PEs	256x256	512x64	512x512
Syst. Array Chiplets Num.	16	8	64
Scratchpad Chiplets Num. (3D)	16	27	20
Accumulator Chiplets Num.	4	1	8
Embeddings Chiplets Num.	0	0	8
BW [TB/s] per Chiplet	1	1	1
Data lanes per Chiplet	8192	8192	8192
Chiplet Grid	6x6	6x6	10x10
Interposer Size* [mm x mm]	38x38	38x38	64x64

S4: Three different SiPs for three dataflows

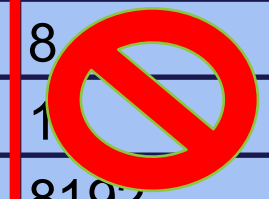
	Sub 20nm SiP		
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Chiplet Grid	6x6	6x6	10x10
Interposer Size* [mm x mm]	38x38	38x38	64x64

S4: Three different SiPs for three dataflows

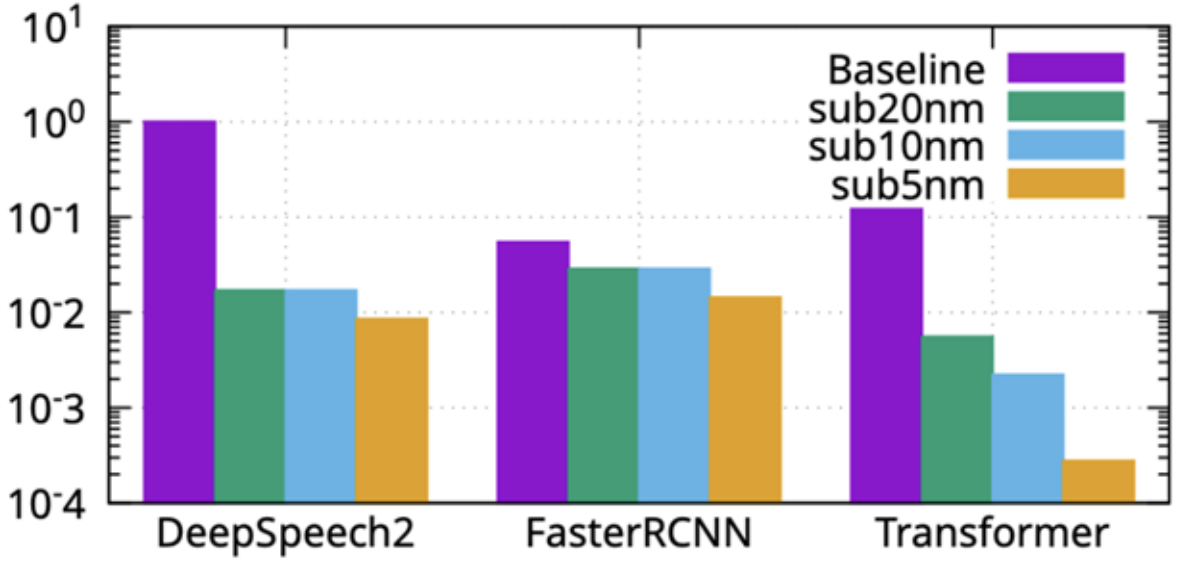
	Sub 20nm SiP		
	CNN	Deep Speech	Trans.
Syst. Array PEs	256x256	512x64	512x512
Syst. Array Chiplets Num.	16	8	64
Scratchpad Chiplets Num. (3D)	16	27	20
Accumulator Chiplets Num.	4	1	8
Embeddings Chiplets Num.	0	0	8
BW [TB/s] per Chiplet	1	1	1
Data lanes per Chiplet	8192	8192	8192
Chiplet Grid	6x6	6x6	10x10
Interposer Size* [mm x mm]	38x38	38x38	64x64

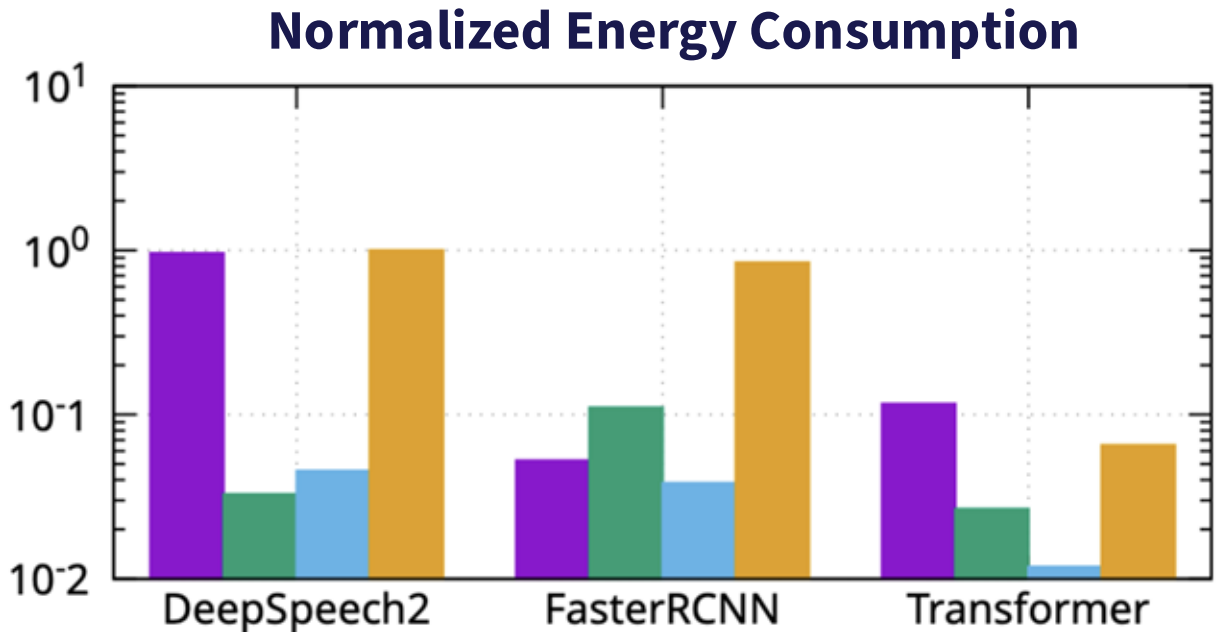


S4: Three different SiPs for three dataflows

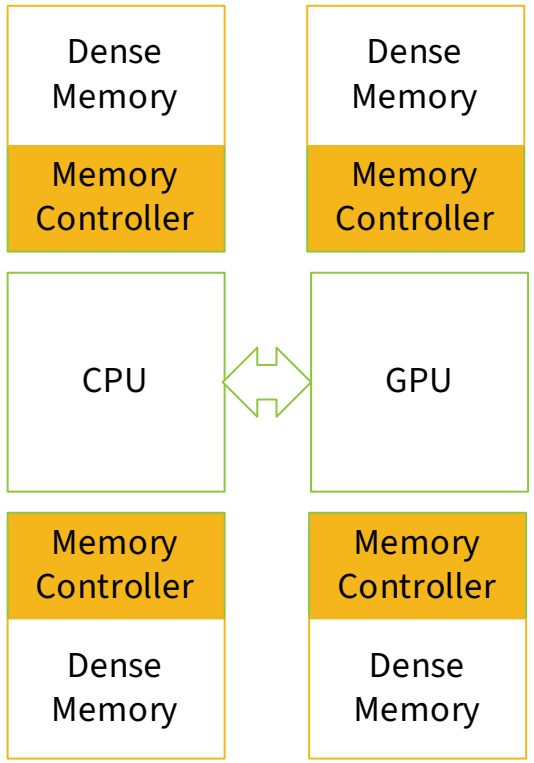
	Sub 20nm SiP			Sub 10nm/5nm SiP		
	CNN	Deep Speech	Trans.	CNN	Deep Speech	Trans.
Syst. Array PEs	256x256	512x64	512x512	256x256	512x128	512x512
Syst. Array Chiplets Num.	16	8	64	4	8	16
Scratchpad Chiplets Num. (3D)	16	27	20	9	21	
Accumulator Chiplets Num.	4	1	8	2	1	2
Embeddings Chiplets Num.	0	0	8	0	0	1
BW [TB/s] per Chiplet	1	1	1	2/4	2/4	2/4
Data lanes per Chiplet	8192	8192	8192	16384	16384	16384
Chiplet Grid	6x6	6x6	10x10	3x3	6x5	6x5
Interposer Size* [mm x mm]	38x38	38x38	64x64	30x28	30x28	30x28

Evaluation: Better Latency and Better Energy consumption

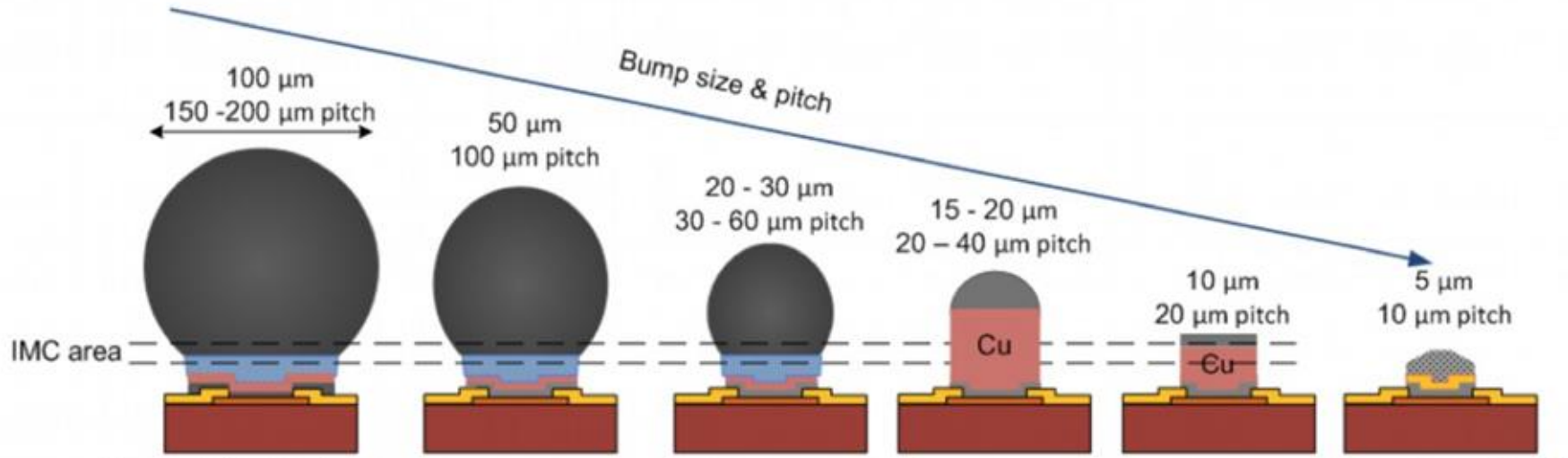




Call to Action: Let's imagine the future together



Today standards

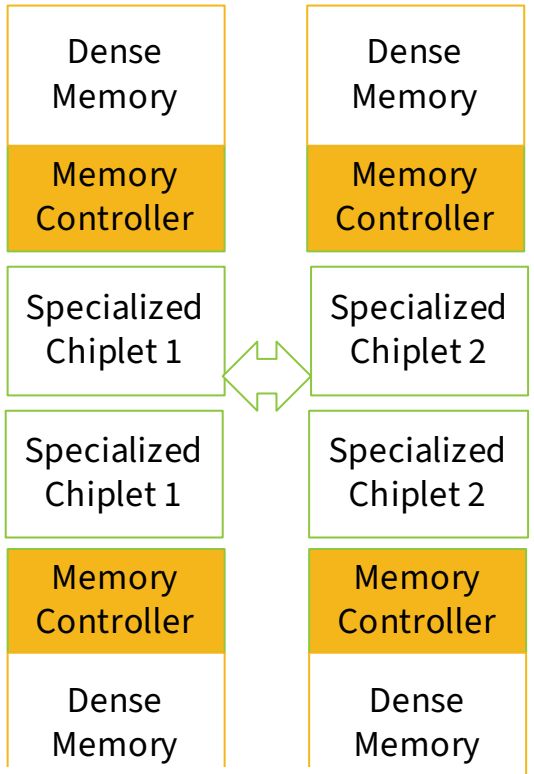


~2024
40µm

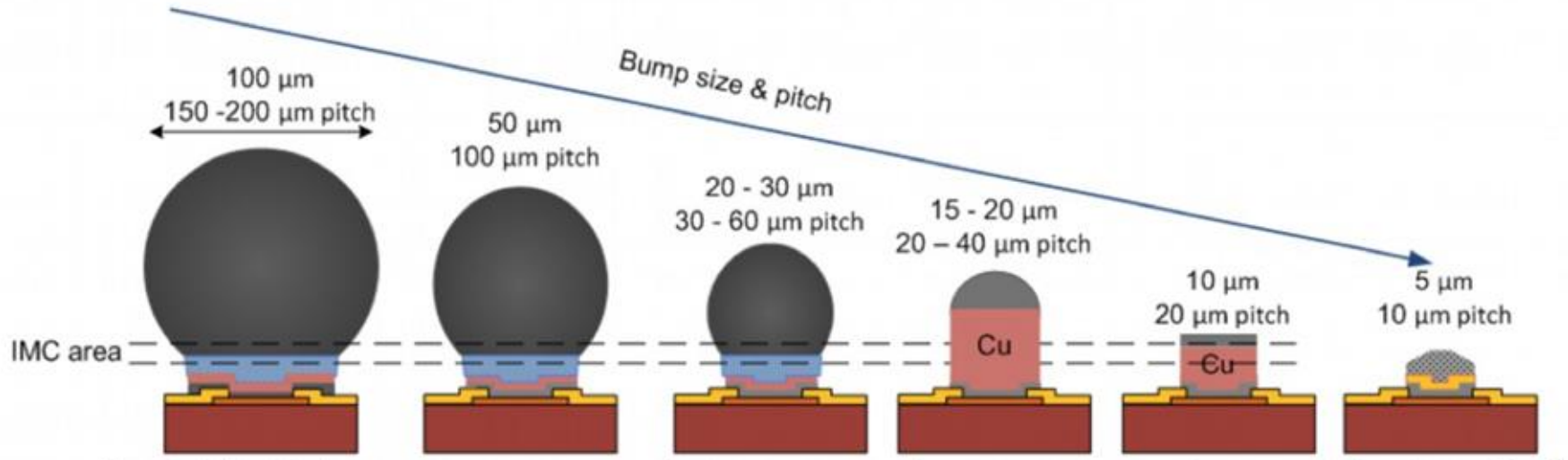
…UCIe
BoW…

Technology moves fast!

Call to Action: Let's imagine the future together



How should the future look like?
Hopefully bright!



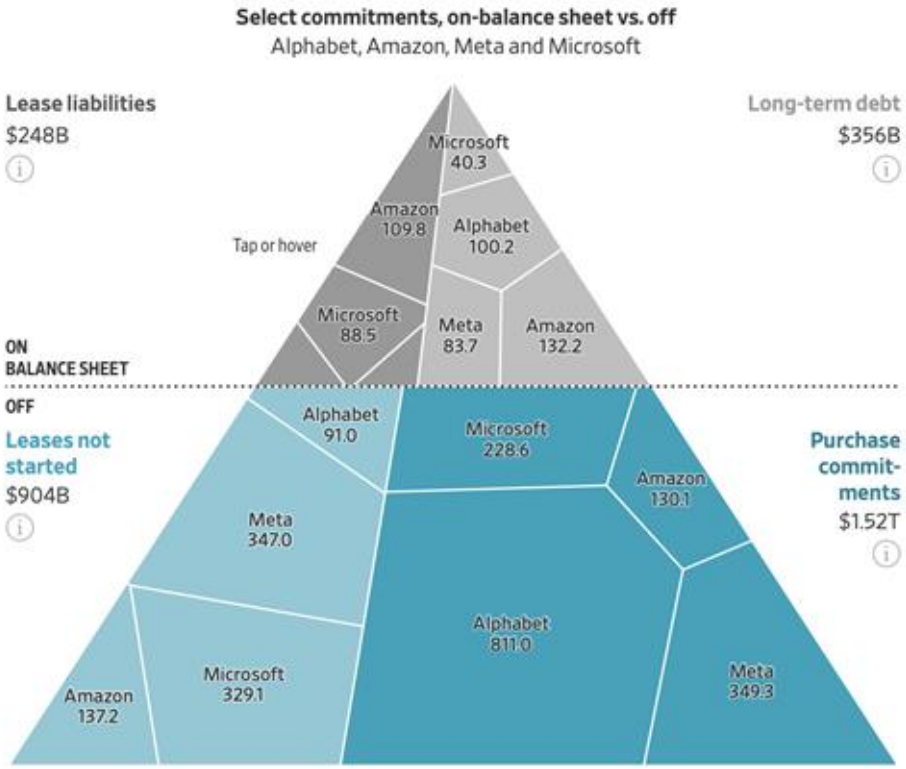
Intels' 25µm

TSMC's 10µm

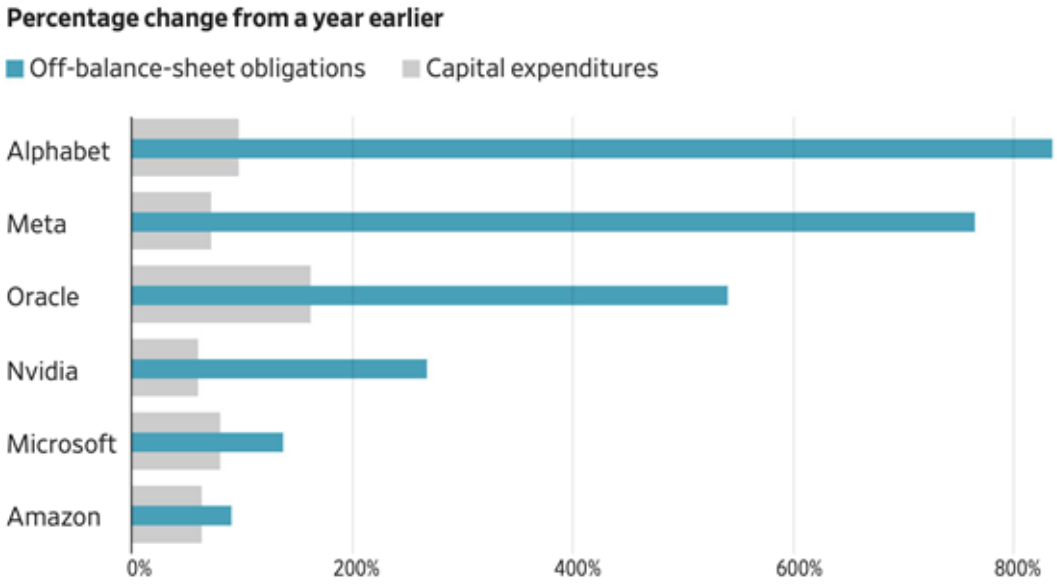
What we talked about 2 years ago is not valid anymore!

THE WALL STREET JOURNAL.

Why Big Tech's AI Spending Is $3 Trillion Higher Than It Seems



Note: Data are as of each company's most recent quarterly filing and in billions, except where noted.
Source: WSJ analysis of company filings

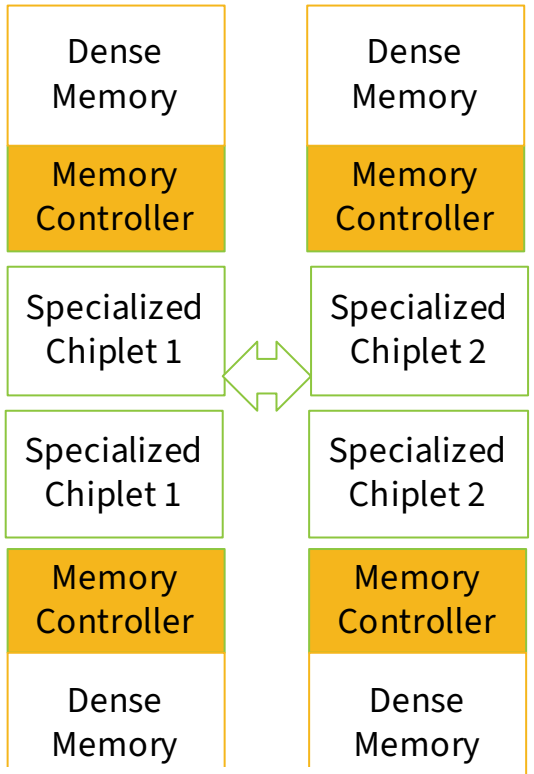


Note: Changes in off-balance-sheet obligations are based on the combined totals of leases not started and purchase commitments. Capex growth is based on trailing 12-month expenditures. All data are as of June except Nvidia, which are as of April, and Meta, which are as of July.
Source: WSJ analysis of company filings

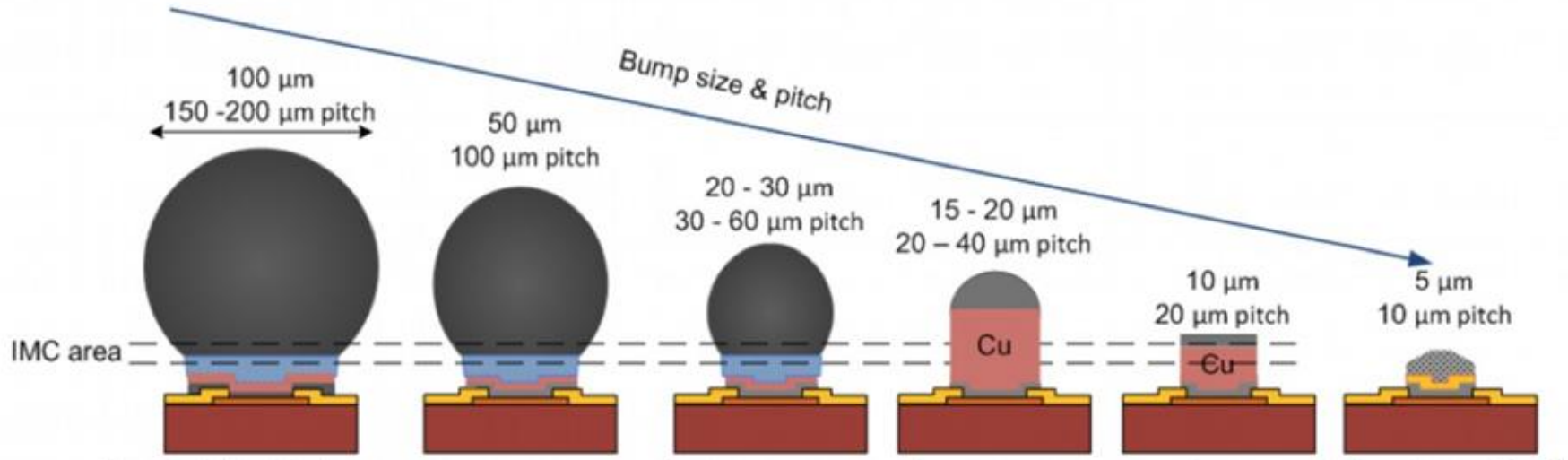
Super Intelligence

The Genesis Mission is a defining moment in American history, marshalling the full resources of our nation to secure technological leadership in the age of Super Intelligence (SI). This initiative combines DOE's and the National Labs' world-class supercomputing power, unique scientific data, and SI capabilities into a single integrated system designed to dramatically accelerate scientific discovery. As strategic competitors race to dominate SI, the Genesis Mission ensures the United States will not only keep pace but will lead, accelerating breakthroughs in energy independence, national security, and critical technologies that will shape the future for generations to come. Learn more at the Genesis Mission website.

Call to Action: Let's imagine the future together



How should the future look like?
Hopefully bright!



Intels' 25µm

TSMC's 10µm

What we talked about 2 years ago is not valid anymore!